MOUHAMADOU FADEL DIOP

Genomic Epidemiology · Transmission Inference · Simulation-Based Methods



PROFILE

Final-year PhD candidate (LSHTM / MRC Unit The Gambia at LSHTM) working on the identifiability of pathogen transmission inference: determining when genomic and epidemiological data contain enough information to recover recent ancestry, transmission linkage and geographic source, at what resolution and with what uncertainty. Experienced in designing forward genetic simulations to validate genomic relatedness measures against known ground truth in *Plasmodium falciparum* and *Chlamydia trachomatis*, and in applying genomic and epidemiological evidence to residual malaria in The Gambia and Senegal. Proficient in R and Python with reproducible, version-controlled workflows. Seeking to extend this work into formal likelihood-based and Bayesian inference of transmission networks.

RESEARCH EXPERIENCE

2022 | Present

PhD Researcher — Genomic Epidemiology & Transmission Inference

London School of Hygiene & Tropical Medicine / MRC Unit The Gambia at LSHTM

London, UK & Banjul, The Gambia

Thesis: Towards Biologically Informed Genomic Transmission Inference: A Generalised Framework for Evaluating Transmission Identifiability Across Infectious Diseases. Submission expected November 2026.

- **Framework:** Framed genomic transmission inference as an identifiability problem: which target (recent ancestry, transmission linkage, geographic source) can be recovered from the available observations, at what resolution and with what uncertainty, before methods are compared or used operationally.
- ***P. falciparum* pedigree identifiability:** Built controlled forward simulations to quantify how recombination, population connectivity, sampling, choice of relatedness measure and target definition affect discrimination of recent pedigree classes.
- ***C. trachomatis* transmission linkage:** Simulated ocular *C. trachomatis* transmission to test whether IBD, IBS and cophenetic distance discriminate true transmission-linked pairs from genetically related non-linked pairs. Used simulated true IBD to separate limits of genomic measurement from limits of the underlying evolutionary signal.
- **Geographic source of residual malaria:** Combined genomic and epidemiological evidence to evaluate the sources of residual malaria in western Gambia, estimating support and uncertainty for local acquisition, movement from eastern Gambia and cross-border acquisition in Senegal over time.
- **Validation & software:** Evaluated inference performance against simulated ground truth (ROC, precision-recall/AUPRC). Developed reproducible simulation and analysis workflows in R and Python (SLiM, msprime, tskit, Snakemake), version-controlled on GitHub.

CONTACT INFO

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github.com/mfdiop
LinkedIn
Banjul, The Gambia

ORCID
Google Scholar
ResearchGate

CORE EXPERTISE

Genomic transmission inference
Identifiability & simulation-based validation
P. falciparum & *C. trachomatis* genomic epidemiology
Genomic relatedness (IBD, IBS, phylogenetic distance)
Integration of genomic & epidemiological data
Statistical evaluation (ROC, AUPRC)
Reproducible workflows in R & Python
Linux / HPC

2019 | 2022

Bioinformatician
MRC Unit The Gambia at LSHTM — Banjul, The Gambia

- Processed and quality-controlled sequencing data and associated sample metadata for research projects (alignment, QC, variant-level data in FASTQ/BAM/VCF formats).
- Built standardised, documented workflows that improved the reproducibility and traceability of analyses and data records.
- Prepared analysis-ready datasets, summaries and reports for research teams and project decision-making.

2018 | 2019

Bioinformatics Intern
MRC Unit The Gambia at LSHTM — Banjul, The Gambia

- Cleaned, organised and quality-checked biological data and metadata used for downstream analysis and reporting.

PUBLICATIONS

My full and up-to-date list of peer-reviewed publications is available on ORCID, Google Scholar and ResearchGate. Please use these links to view my publications.

EDUCATION

2022 | Present

PhD in Bioinformatics
London School of Hygiene & Tropical Medicine, London, UK

Thesis submission expected November 2026.

2014

MSc in Bioinformatics
Cheikh Anta Diop University, Dakar, Senegal

2011

BSc in Bioinformatics-Biomathematics
Cheikh Anta Diop University, Dakar, Senegal

2010

BSc in Physics and Chemistry
Cheikh Anta Diop University, Dakar, Senegal

TEACHING & COMMUNICATION

- Delivered introductory training on data handling and reproducible R workflows.
- Communicated computational and genomic results to interdisciplinary teams through reports and visualisations.

Technical Skills

Simulation & population genetics

SLiM (forward-time), msprime, tskit
IBD, IBS, phylogenetic / cophenetic distance

Statistics & evaluation

Simulation-based validation
ROC, precision-recall (AUPRC)

Programming

R (tidyverse, Bioconductor, Shiny, R Markdown)
Python

NGS processing

BWA, Minimap2, SAMtools, GATK, Qualimap
FASTQ, BAM, VCF

Workflows & systems

Snakemake, Git / GitHub
Linux / HPC

Languages

Wolof: Native
French: Proficient
English: Proficient

Last updated on 2026-10-05.